

THE DHYM EQUATION ON SURFACES: A POSITIVITY CRITERION AND ITS MECHANISM

ABSTRACT. On a compact Kähler surface, positive-phase scalar dHYM solvability is an ordinary Kähler-cone condition after a cohomological shift. We state the full criterion and derive its reduction to the complex Monge–Ampère equation, including the converse phase check. This is a surface-focused expository sample, not a comprehensive literature update.

1. INTRODUCTION

Fix a Kähler form ω on a compact complex surface X . For a smooth real $(1, 1)$ -form η , let λ_1, λ_2 be its ω -eigenvalues. With each arctangent in $(-\pi/2, \pi/2)$,

$$(\omega + \sqrt{-1}\eta)^2 = \prod_{j=1}^2 (1 + \sqrt{-1}\lambda_j) \omega^2, \quad \vartheta_\omega(\eta) = \sum_{j=1}^2 \arctan \lambda_j.$$

For a fixed closed representative $\alpha \in A$, the unknown is a real smooth potential φ , with $\eta = \alpha + \sqrt{-1}\partial\bar{\partial}\varphi$. The scalar dHYM equation is $\vartheta_\omega(\eta) = \hat{\theta}$. The positive phase fixes the lift, not merely the phase modulo π . Here $D \cdot E = \int_X D \wedge E$ and $\mathcal{K}_X$ is the Kähler cone.

Theorem 1.1. *Let X be a compact Kähler surface, $\Omega \in \mathcal{K}_X$, and $A \in H^{1,1}(X, \mathbb{R})$ with $s = A \cdot \Omega > 0$. Set*

$$\hat{\theta} = \text{Arg} \int_X (\Omega + \sqrt{-1}A)^2 \in (0, \pi), \quad c = \cot \hat{\theta} = \frac{\Omega^2 - A^2}{2s}.$$

The following are equivalent:

- (i) *For every Kähler form $\omega \in \Omega$, there is a smooth closed real $(1, 1)$ -form $\eta \in A$ such that $\sum_{j=1}^2 \arctan \lambda_j(\eta; \omega) = \hat{\theta}$.*
- (ii) *$B := A + c\Omega$ is a Kähler class.*
- (iii) *$B \cdot C > 0$ for every irreducible curve $C \subset X$.*

The analytic reduction is due to Jacob–Yau [JY17]; the last condition uses the numerical description of the surface Kähler cone [DP04]. The strict curve inequality is essential: nefness is not the asserted solvability criterion.

2. THE SHIFT AND THE MONGE–AMPÈRE EQUATION

Theorem 1.1 (recalled). *Let X be a compact Kähler surface, $\Omega \in \mathcal{K}_X$, and $A \in H^{1,1}(X, \mathbb{R})$ with $s = A \cdot \Omega > 0$. Set*

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- (ii) $B := A + c\Omega$ is a Kähler class.
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Write $\alpha_j = \arctan \lambda_j$ and $\beta = \eta + c\omega$. If $\alpha_1 + \alpha_2 = \widehat{\theta} \in (0, \pi)$, then

$$\lambda_j + c = \frac{\cos(\widehat{\theta} - \alpha_j)}{\sin \widehat{\theta} \cos \alpha_j} > 0.$$

Indeed $\widehat{\theta} - \alpha_j$ is the other angle in $(-\pi/2, \pi/2)$. Thus $\beta > 0$. Expanding the imaginary part gives

$$\begin{aligned} \operatorname{Im}(e^{-\sqrt{-1}\widehat{\theta}}(\omega + \sqrt{-1}\eta)^2) = 0 &\iff \eta^2 + 2c\omega \wedge \eta = \omega^2 \\ &\iff \beta^2 = (1 + c^2)\omega^2. \end{aligned}$$

In particular, a solution supplies a Kähler representative of B .

Conversely, suppose $B \in \mathcal{K}_X$ and choose $\beta_0 \in B$ positive. Cohomological compatibility is automatic:

$$B^2 = A^2 + 2cs + c^2\Omega^2 = (1 + c^2)\Omega^2.$$

Yau's theorem [Yau78] produces a smooth ψ , unique modulo constants, with

$$\beta = \beta_0 + \sqrt{-1}\partial\bar{\partial}\psi > 0, \quad \beta^2 = (1 + c^2)\omega^2.$$

Let $\eta = \beta - c\omega$. Its class is A , and its eigenvalues satisfy $(\lambda_1 + c)(\lambda_2 + c) = 1 + c^2$ with both factors positive. Hence

$$\lambda_1 + \lambda_2 \geq 2\sqrt{1 + c^2} - 2c > 0.$$

Therefore $\operatorname{Im}(\omega + \sqrt{-1}\eta)^2 > 0$. Together with the vanishing imaginary part after rotation, this selects $\vartheta_\omega(\eta) = \widehat{\theta}$, not another lift. The $\partial\bar{\partial}$ -lemma expresses η in the prescribed potential class. This proves the first two conditions equivalent.

The numerical step also needs the correct component of the positive cone. By the Hodge index inequality, $A^2\Omega^2 \leq s^2$, so

$$B \cdot \Omega = \frac{2s^2 + \Omega^4 - A^2\Omega^2}{2s} \geq \frac{s^2 + \Omega^4}{2s} > 0, \quad B^2 > 0.$$

On a compact Kähler surface, a real $(1, 1)$ -class with these two properties is Kähler precisely when it pairs positively with every irreducible curve [DP04]. The component condition has thus been checked rather than suppressed.

3. A MODEL AND THE BOUNDARY OF THE COMPARISON

If $A = a\Omega$ with $a > 0$, choose $\eta = a\omega$. Then

$$\widehat{\theta} = 2 \arctan a, \quad c = \frac{1 - a^2}{2a}, \quad B = \frac{1 + a^2}{2a}\Omega \in \mathcal{K}_X.$$

This model displays both the cohomological shift and its strict positivity. For a general class, testing A alone would miss the phase-dependent term. Neither this calculation nor

the existence theorem proves convergence of an evolution starting at an arbitrary initial potential.

The mirror-motivated organizing question is the Thomas–Yau problem [TY02]. For a Calabi–Yau metric with normalized parallel holomorphic volume form Y , a graded Lagrangian L has a real phase lift θ_L defined by $Y|_L = e^{\sqrt{-1}\theta_L} dV_L$. A special Lagrangian has constant θ_L . The existence question asks which appropriate stability conditions select Hamiltonian classes containing such a representative. The flow question additionally asks what evolution and treatment of singularities reach that representative. These are different assertions; Joyce’s categorical and surgery program [Joy15] is not a theorem of unconditional smooth convergence.

The surface theorem above is complete in its stated scalar scope. It does not identify general categorical stability with dHYM solvability, settle Thomas–Yau, or extend the surface curve test to arbitrary dimension. Those threads belong to the full reference article rather than to a compressed statement with hidden hypotheses.

REFERENCES

- [DP04] J.-P. Demailly and M. Păun, *Numerical characterization of the Kähler cone of a compact Kähler manifold*, Ann. of Math. **159** (2004), 1247–1274.
- [JY17] A. Jacob and S.-T. Yau, *A special Lagrangian type equation for holomorphic line bundles*, Math. Ann. **369** (2017), 869–898; [arXiv:1411.7457](#).
- [Joy15] D. Joyce, *Conjectures on Bridgeland stability for Fukaya categories of Calabi–Yau manifolds, special Lagrangians, and Lagrangian mean curvature flow*, EMS Surv. Math. Sci. **2** (2015), 1–62; [arXiv:1401.4949](#).
- [TY02] R. P. Thomas and S.-T. Yau, *Special Lagrangians, stable bundles and mean curvature flow*, Comm. Anal. Geom. **10** (2002), 1075–1113; [arXiv:math/0104197](#).
- [Yau78] S.-T. Yau, *On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation, I*, Comm. Pure Appl. Math. **31** (1978), 339–411.